STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA
DIPLOMA EXAMINATION (C-24)
C-24-DEC-2025
SEMESTER III , SEMESTER END EXAM
AI/CCB/CPS/CS

CS-305

Computer Organization And Architecture

PCODE 243032

Exam Date: 04-12-2025

Duration: 2 Hours [10:00 AM To 12:00 NOON]

Session: FN

[Total Marks: 40]

PART-A

Instructions: 1. Answer the following questions. 8 X 1 = 8
2. Each question carries **ONE** mark.

1. What are the operations used in performing fixed point division?
2. Define Opcode .
3. List out any three Input devices.
4. What is meant by Access time ?

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5. What is interface?
6. Define Synchronous data transfer.
7. What is Symmetric Multi Processor (SMP)?
8. What is the function of a GPU?

PART-B

Instructions: 1. Answer the following questions. 4 X 3 = 12
2. Each question carries **THREE** marks.

9(a). Multiply (-6) and (2) using Booth's algorithm.

----- OR ----

9(b). What is Programmed I/O method of data transfer?

10(a). Explain the registers essential to instruction execution.

----- OR ----

10(b). Interpret the concept of a multi-core processor.

11(a). Write about any three Features of Interrupt driven input/output.

----- OR ----

11(b). Explain about the strobe controlled procedure of data transfer.

12(a). List three advantages of Symmetric Multi Processors (SMP).

----- OR ----

12(b). Differentiate Between RISC and CISC Processors.

PART-C

Instructions: 1. Answer the following questions. 4 X 5 = 20

2. Each question carries **FIVE** marks.

13(a). Explain the functional block diagram of digital computer with diagram.

----- OR ----

13(b). Explain about DMA Controlled transfer.

14(a). Explain Zero address, one address, two address and three address instruction with examples.

----- OR ----

14(b). Compare Central Processing Unit and Graphics Processing Unit.

15(a). Write about handshaking procedure of data transfer.

----- OR ----

15(b). Explain Input Output Processor with neat diagram.

16(a). Compare and contrast scalar processing and parallel processing along with advantages and limitations.

----- OR ----

16(b). Explain the parallel processing principles.